Literature Review & Study Design

Bao T. Truong · 05.11.2026

Technische Universität Dresden

SynoSys

Objectives

- Trace a study design
- Identify the theory, hypothesis, recruitment, randomization, measurement, analysis, and interpretation
- Know how to identify research topic independent of the methods

In-class student-led discussion

Paper/RQs	Treatment & comparator	Concept - > operationalization	Outcome variable	Conclusion	Strict or naturalistic setting
Debunking misinfo on Tiktok with videos	Conditions - Misinfo videos - Debunking videos - Correction videos On 6 topics	- Truth discernment - Belief in false claim	(1) Ability to tell truth from false (accuracy, reliability, impartiality) — rating from 0-100 (2) Belief for false claim — scale of 1-7	- Debunking increase truth discernment - Correction with context helps with outcome (1)	- Strict setting
Instagram inspiration How can content positively inspire a user??	Conditions - Exposure to nature photos (e.g., scenic photos) - Baseline?	Inspiration	Attitude towards the author 1-10 Inferior / Superior Personal well-being PANAS items	Positive effect of nature content in inspiring people (with the caveat that a person can feel envious)	- Strict setting

Some concerns raised by students:

- Durability & generalizability of the observed effects
- ignore too many other effects

New terms:
Pre-registered
Generalizability

Tiktok paper described by Antonia and Anna-Theresa
Instagram paper described by David

Technische Universität Dresden

SynoSys

Independent variables X ? Y Dependent variables

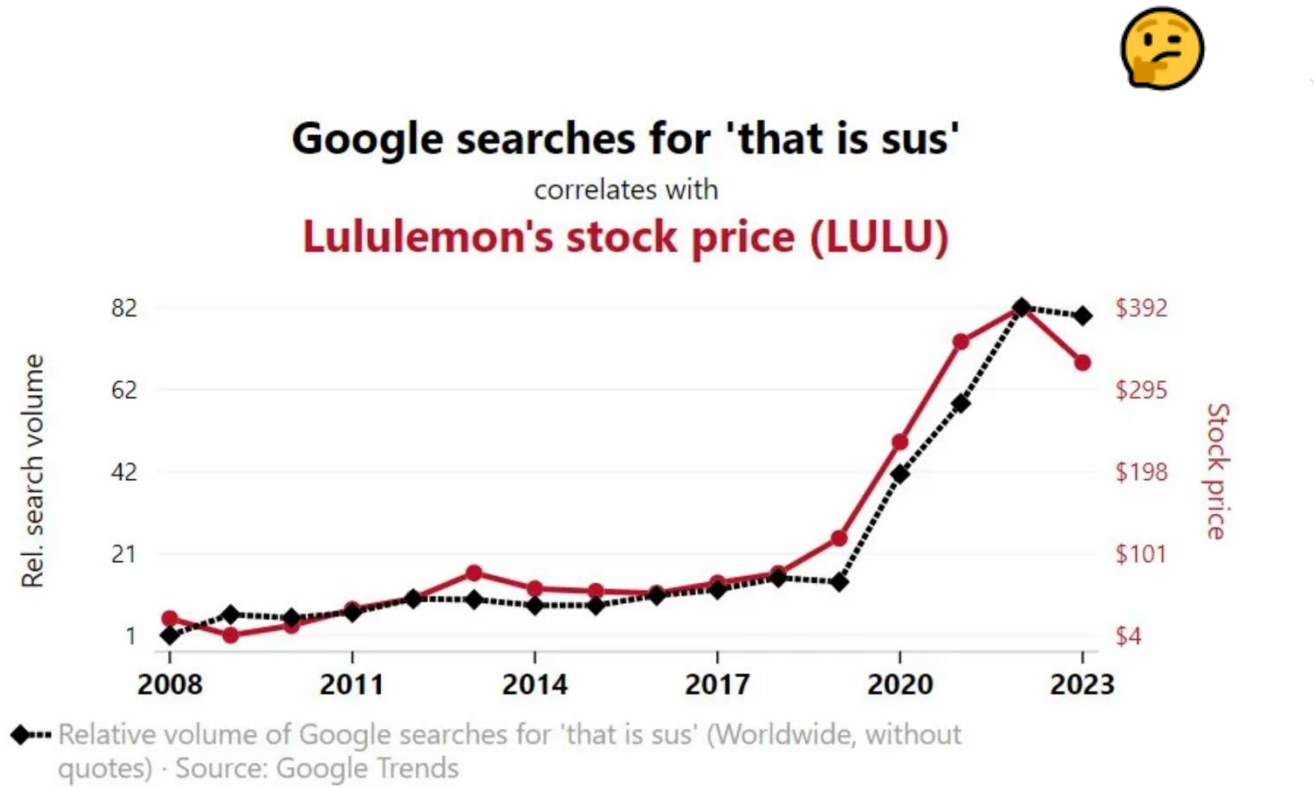
Some type of intervention, e.g.,

- A way of ranking
- Some type of content
- Some re-bunking

Outcomes

- Mental health
- Hate speech
- Misinformation
- Polarization
- Steoreotypes
- Political participation
- Increase in knowledge
- Individual expression

Correlation is not Causation



$$Y = \beta X + u$$

Or LULU stockprice = $\beta Google\ search + u$

- β here is estimated from observational data.
- This means that X is not independent from unobserved causes that affect Y.
  Something else, like time or internet culture, may be responsible for the rise of both quantities
- The estimation β, then, combines the effect of X

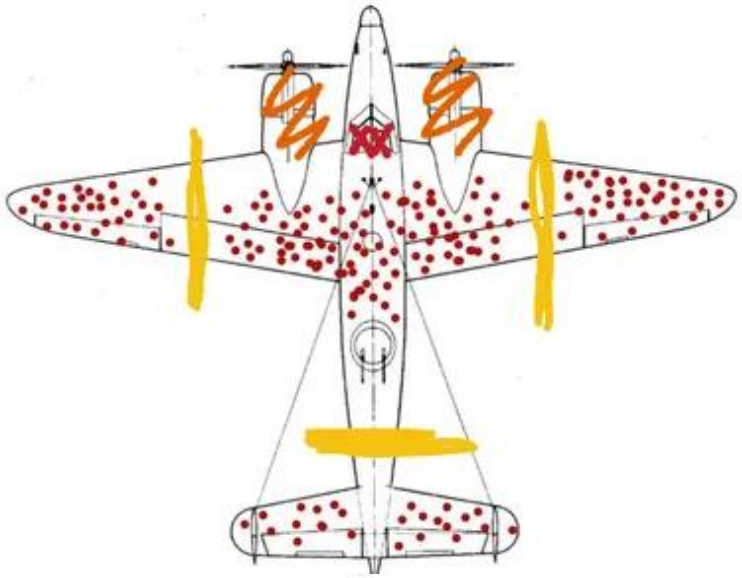
Unless there's a way to attribute

https://tylervigen.com/spurious-correlations

Selection bias

Happens when the sample or treatment choice depends on factors related to outcomes

🤔?





Randomization

Proper randomization accounts for the source of variation in X, allowing us to make causal claims.

e.g., Suppose we want to estimate the causal effect of wing reinforcement (placing extra armor on the wings — this is the treatment) on survival probability.

We'd need to randomly assign planes to extra armors* before the mission, then tracking the faith of all planes.

It's crucial here that the observation is made for all planes.

*In reality, randomizing potentially life-critical armor configurations would be ethically and operationally difficult.

Technische Universität Dresden

SynoSys

Back to our Lululemon example

- With just common sense, it's hard to believe that the surge in the search term is the cause for stock price rise.
- In principle, we can *try* to estimate causation by the following:
  - Randomly expose some people or regions to a campaign that induces searches for "that's sus," while others are not exposed (this is the treatment)
  - Measure whether this induced increase in searches changes a well-defined outcome*
  - Ensure the treatment itself has no other route to affect the outcome.

Stock price is not a well-defined outcome, as everyone shares the same price, and people's treatments can affect one another through financial markets.
Even assuming that the treatment (randomly increasing searches in one region) might move the global stock price; this might also change sentiment or purchases directly, so it would not isolate the effect of searches.

Briefly about DAG (Directed Acyclical Graph)

- A powerful tool for representing and reasoning about causal relationships
- DAGs capture the directional flow of causality
- Nodes (boxes) represent entities of interests
- Edges (arrows) represent immediate cause-and-effect relationship between variables
- No loops/cycles (hence “acyclic”) — the absence of them reflects the unidirectional nature of time and causation

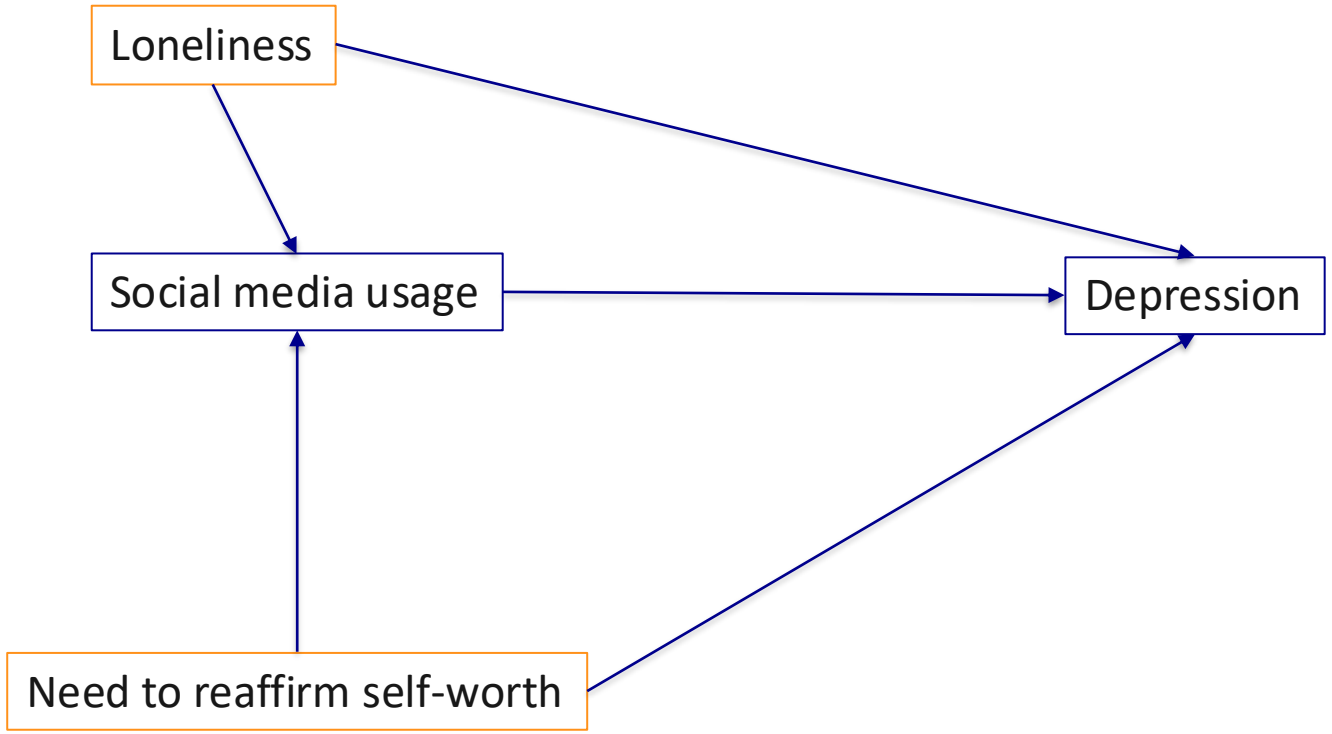
Suppose we want to study the causal effect of Social media usage on depression.
This is what it looks like as a DAG



More about DAGs in this book chapter -> https://mixtape.scunning.com/03-directed_acyclical_graphs#introduction-to-dag-notation
This book (free) is very useful to learn about causal inference

Technische Universität Dresden | SynoSys

Let’s expand our DAG a bit more



Confounder

A common cause of the treatment and outcome that creates a non-causal association between them (a backdoor path)

Collider

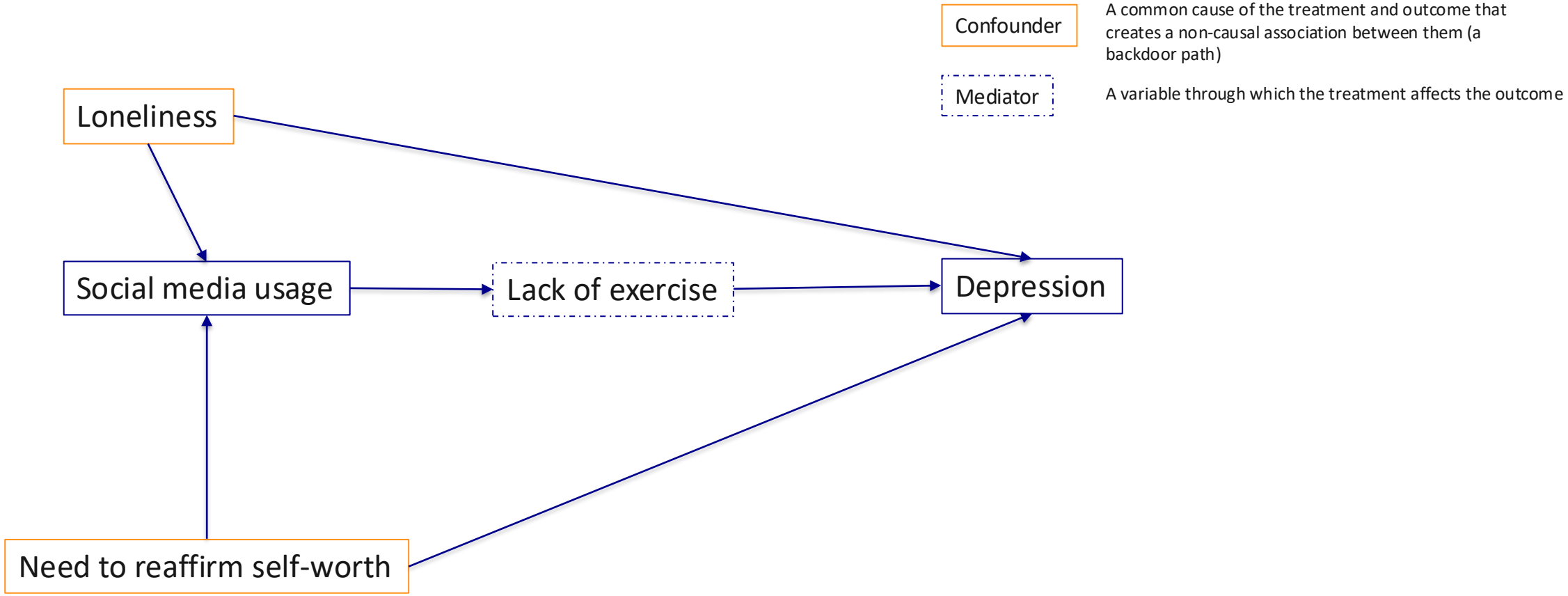
A common effect of the treatment and outcome.

More here: https://mixtape.scunning.com/03-directed_acyclical_graphs#introduction-to-dag-notation
And here, if you prefer videos: https://youtu.be/7lPWtFXsuzk

Technische Universität Dresden

SynoSys

Let’s expand our DAG a bit more

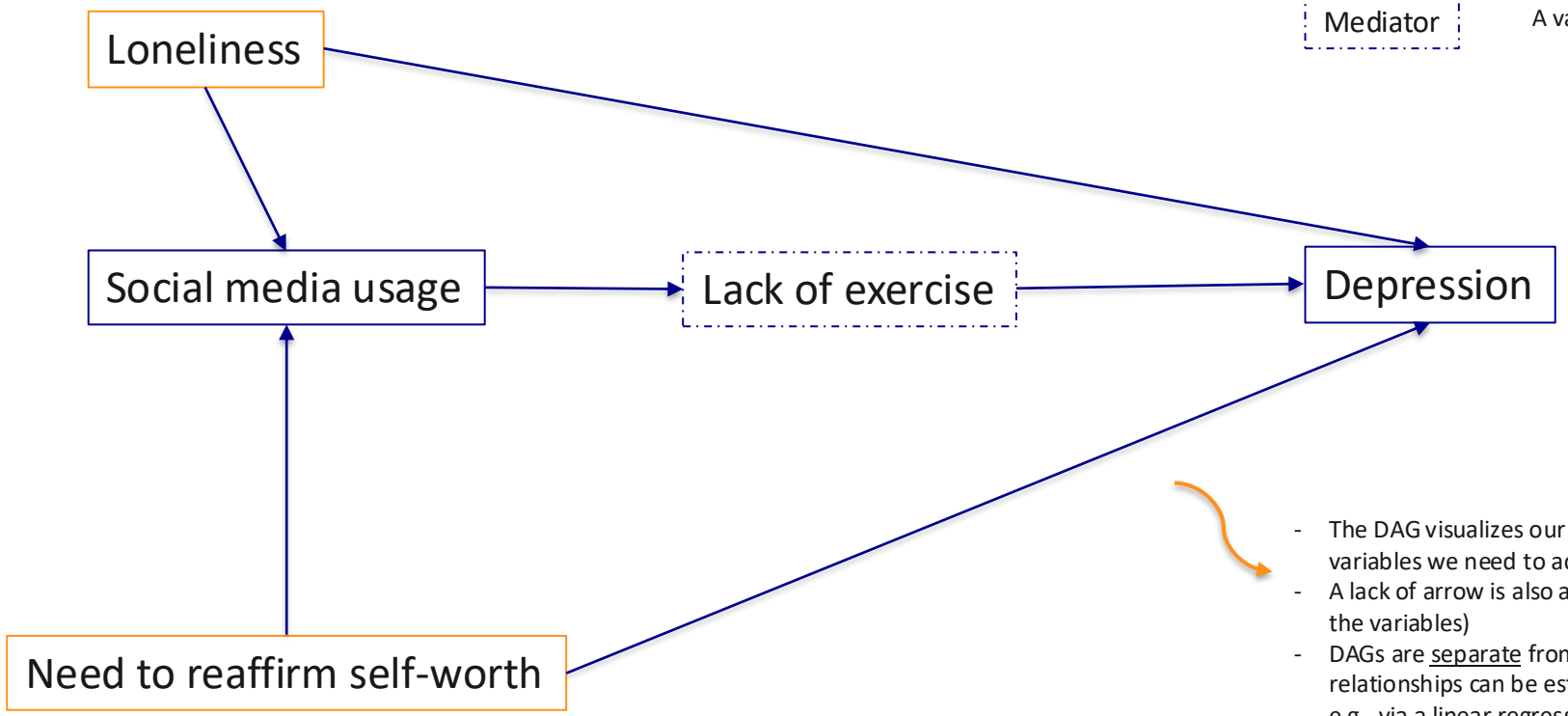


More here: https://mixtape.scunning.com/03-directed_acyclical_graphs#introduction-to-dag-notation
And here, if you prefer videos: https://youtu.be/7lPWtFXsuzk

Technische Universität Dresden

SynoSys

Let’s expand our DAG a bit more



Confounder — A common cause of the treatment and outcome that creates a non-causal association between them (a backdoor path)

Mediator — A variable through which the treatment affects the outcome

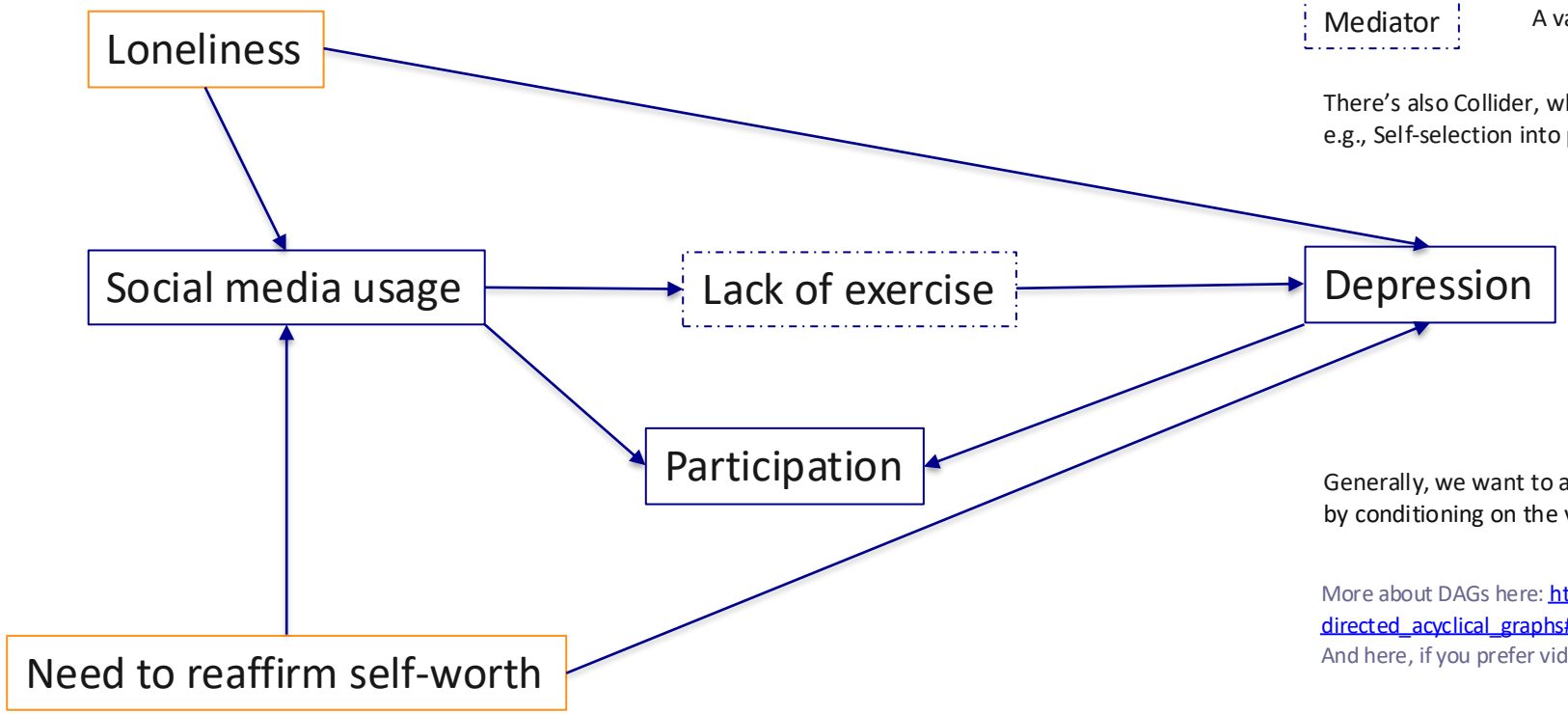
- The DAG visualizes our theory & helps as a mental model for all the variables we need to account for
- A lack of arrow is also an assumption (that there’s no relationship between the variables)
- DAGs are separate from the functional form of the estimation — the causal relationships can be estimated in different ways
- e.g., via a linear regression:
- or polynomial regression

More here: https://mixtape.scunning.com/03-directed_acyclical_graphs#introduction-to-dag-notation
And here, if you prefer videos: https://youtu.be/7lPWtFXsuzk

Technische Universität Dresden

SynoSys

Let’s expand our DAG a bit more



Confounder — A common cause of the treatment and outcome that creates a non-causal association between them (a backdoor path)

Mediator — A variable through which the treatment affects the outcome

There’s also Collider, which is a common effect of the treatment and outcome. e.g., Self-selection into participation is a collider.

Generally, we want to a) leave the mediators alone; b) block backdoor paths by conditioning on the variable; and c) don’t condition on a collider

More about DAGs here: https://mixtape.scunning.com/03-directed_acyclical_graphs#introduction-to-dag-notation
And here, if you prefer videos: https://youtu.be/7IPWtFXsuzk

How to develop a research topic

A good research topic is one that:

- Strongly interests you
- Potentially interests your peers and the scientific community

(Hint: Look for gaps in the literature. Has it been studied before?)

- Not to broad

correctly

(Hint: Can it fit into a five-pager? Does it have enough context for a person looking to do a similar study?)

“Does social media cause polarization?” — too broad & not answerable

exposure to politician posts on X

Gap in feelings between partisans (affective polarization)
Gap in issue stance between partisans (ideological polarization)
Or something else, e.g., distance in the network

Additional context

Which population?
German voters
Heavy social media users
Adolescent

When?
During the elections
During Christmas

Narrower scope!

1. What are the effects of exposure to politician posts on X on German voters?
2. Among regular users of platform X, does reducing exposure to politically hostile misinformation for 14 days improve out-party warmth and misinformation discernment without reducing news use or perceived autonomy?

Scoping the question

1. What are the effects of exposure to politician posts on X on German voters?

2. Among regular users of platform X, does reducing exposure to politically hostile misinformation for 14 days improve out-party warmth and misinformation discernment without reducing news use or perceived autonomy?

Exploratory

Confirmatory

This is a hypothesis about the directional relationship between an independent variable (exposure to politically hostile misinformation) and dependent variables (out-party warmth, misinfo discernment, news use, perceived autonomy)

Reliability, Validity & Generalizability

- Reliability: the measurement/procedure produces consistent results under the same conditions (results are repeatable)
- Validity: the study measures & inference accurately the quantities it claims to study
- Generalizability: the finding is likely to apply beyond the particular study sample (to other settings and time periods)

Types of validity

- Internal validity: Is the observed difference plausibly caused by the intervention rather than other alternative explanations? (e.g., confounding, attrition)
- External validity: Can we see the same results in samples with similar characteristics?
- Ecological validity: Does the study resemble real-world settings & behaviors?
- Construct validity: Does the operational measure match the concept? (e.g., does a feeling thermometer adequately capture affective polarization?)
- Statistical validity: Do the data and analysis support the statistical conclusion?

Trade-offs between internal vs external validity

INTERNAL VALIDITY

EXTERNAL VALIDITY



Burton, Herzog & Lorenz-Spreen (2026)

Roozenbeek et al. (2022) covers both

Piccardi et al. (2025)

Studies often need to balance between internal, external and ecological validity

Technische Universität Dresden

SynoSys

Study design workflow

Semester 1

1. Identify the topic
2. Come up with research question(s)
3. Define and operationalize the concepts
4. Choose the study setup

5. Deliver the intervention
6. Measure and analyze
7. Evaluate the claim

Semester 2

- What gap in prior research motivates this study?
- What concepts and mechanism does the study propose?
- How are the concepts translated into observable variables, and how are they measured?
- Is it an experiment, or observational study? What's the sampling, recruitment, and randomization strategy? What does this setup prioritize?

Homework

- Read Burton, Herzog & Lorenz-Spreen (2026) Simple changes to content curation algorithms affect the beliefs people form in a collaborative filtering experiment — trace the study design and identify relevant components (as we did in class)
- Check out Pagnini et al., 2025. Unexpected events and prosocial behavior: the Batman effect